

P2: Manipulation-Aware Refusal Mechanisms

Mid-Project Progress Report

Group 39

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Problem Statement

The Challenge

- LLMs undergo extensive safety fine-tuning to prevent harmful outputs
- Jailbreak attempts manipulate models into generating toxic content
- Conversational manipulation is subtle and hard to detect automatically
- Current systems lack psychologically-aware defenses

MentalManip Dataset

2,915

Dialogues total

11

Manipulation Techniques

5

Victim Vulnerability Types

2

Sets: Control (Dc) and Manipulation (Dm)

Manipulation Techniques (M)

The MentalManip dataset defines 11 manipulation techniques present in dialogues:

1. Denial

5. Playing the Victim Role

9. Brandishing Anger

2. Evasion

6. Playing the Servant Role

10. Accusation

3. Feigning Innocence

7. Shaming or Belittlement

11. Persuasion or Seduction

4. Rationalization

8. Intimidation

What We Built: Manipulation Detector

Component A | RoBERTa Classifier

Base Model	roberta-base (125M parameters, HuggingFace Transformers)
Input	Dialogue text tokenised up to max_length = 128 tokens
Head	Linear classification head added on top of [CLS] token output
Output	Binary label: Manipulation (1) or Normal (0)
Loss	CrossEntropyLoss, Adam optimiser, 1 training epoch on CPU
Data Split	60% train / 20% validation / 20% test (stratified)

Pipeline Files (p2_refusal_pipeline/)

`data_loader.py`

Loads mentalmanip_con.csv, applies 60/20/20 split, returns HuggingFace Dataset objects ready for training.

`train_detector.py`

Fine-tunes roberta-base using the Trainer API. Saves the best checkpoint based on validation loss.

`refusal_policy.py`

Keyword-based classifier that maps detected text to a scenario tag and selects a deterministic response template.

`evaluate.py`

Runs inference on the test set, computes accuracy, precision, recall and F1, and prints the confusion matrix.

`demo.py`

Interactive CLI that accepts user input, runs detection and refusal policy end-to-end, and prints the result.

What We Built: Rule-Based Refusal Policy

1

Manipulation Detected

The RoBERTa classifier flags the incoming user utterance as manipulative (confidence above threshold). Normal utterances bypass the policy entirely and receive a standard assistant response.

2

Scenario Classification

The flagged text is scanned for keywords mapped to scenario tags such as `guilt_trip`, `pressure`, `flattery`, `threat` and `victimhood`. This lightweight keyword match requires no additional model call.

3

Template Response Issued

A deterministic template is selected for the matched scenario. Every template follows a fixed structure: (a) acknowledge the user's feelings, (b) set a clear boundary, (c) offer a constructive alternative path forward.

Key design choice: no free-form generation in the refusal path. Templates guarantee the model cannot be tricked into producing unsafe output through the refusal response itself.

Experimental Results

70.50%

Accuracy

70.73%

Precision

97.25%

Recall

81.89%

F1 Score

Confusion Matrix

	Pred: Normal	Pred: Manip
True: Normal	22	161
True: Manip	11	389

Key Insight

- ✓ 97.25% recall: nearly all manipulation instances are caught
- ✓ Refusal templates fire correctly on detected manipulation
- ✓ Normal queries pass through without triggering refusal
- ! Precision at 70%: some normal utterances flagged as manipulation
- i For a safety system, high recall is prioritised over precision

Team Roles | Group 39

Pranjal Prajapati

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Current Work

- ▶ RoBERTa fine-tuning and training loop
- ▶ Pipeline: data_loader, train, evaluate
- ▶ Demo CLI interface

Next Steps

- > 11-class multi-label classifier
- > LoRA / prefix-tuning adapter
- > Adversarial prompt stress tests

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Current Work

- ▶ Refusal policy design and heuristic rules
- ▶ Template response generation logic
- ▶ Qualitative testing across scenario types

Next Steps

- > Vulnerability detection module
- > Multi-turn context window modelling
- > Human evaluation framework design

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Current Work

- ▶ Dataset analysis and preprocessing
- ▶ Evaluation metrics and confusion matrix
- ▶ Documentation and reporting

Next Steps

- > Annotator-level uncertainty analysis
- > Expanded evaluation suite
- > Dataset utilisation improvements

Plan: Next Steps

Phase 1 | Fine-Grained Classification

Extend binary detector to a 11-class multi-label model using the full MentalManip annotations. Enables technique-specific refusal strategies.

Phase 2 | Vulnerability Modeling

Predict victim vulnerability type from 5 available types. Adapt refusal tone to be more empathetic for dependency cases and firmer for intimidation.

Phase 3 | LLM Adaptation via LoRA

Use the detection signal as a control input for lightweight adapters (LoRA or prefix-tuning) applied to open-source LLMs such as Mistral and LLaMA.

Phase 4 | Multi-Turn Context

Move from single-turn to full conversation history modelling. Detect cumulative manipulation patterns that only become apparent over several dialogue turns.

Phase 5 | Robust Evaluation

Add human evaluation for politeness, helpfulness and resistance. Include adversarial and jailbreak stress tests. Use annotator disagreement for uncertainty.

Phase 6 | Generative Refusals

Replace deterministic templates with a fine-tuned small generative model or a retrieval-based system for more natural and context-sensitive refusal responses.

Summary and Takeaways

What We Achieved

✓ Working detector

RoBERTa fine-tuned on 2,915 dialogues achieves 97.25% recall, catching nearly all manipulation instances.

✓ Safe refusal path

Deterministic template policy with zero free-form generation. The refusal itself cannot be exploited.

✓ Fully local pipeline

No API calls, no cloud cost. Runs entirely on CPU with open-source libraries: PyTorch, HuggingFace, scikit-learn.

✓ Modular codebase

Five clean modules (data_loader, train, refusal_policy, evaluate, demo) that can be swapped or extended independently.

Known Limitations

! Precision at 70%: 161 normal utterances are mis-flagged as manipulation.

! Binary only: technique identity (which of 11 types) is not yet predicted.

! Single-turn only: multi-turn manipulation patterns are not yet captured.

Future Work

- 1 Extend to 11-class classifier to unlock technique-specific refusals (Pranjal)
- 2 Add vulnerability prediction to personalise refusal tone (Roshan)
- 3 Integrate LoRA adapters on a small open-source LLM (Pranjal + Roshan)
- 4 Build expanded evaluation suite with adversarial inputs (Sujal)